WHAT IS DEVOPS

DevOps is a methodology of implementing the automation on each and every stage.

DEVOPS SYLLABUS

1. Git / github / gitlab
2. Docker
3. Kubernetes
4. Terraform
5. Jenkins
6. Datadog

COMPANY:

A company is an organization which is created from two or more people for their profits.

COMPANY DEPARTMENTS

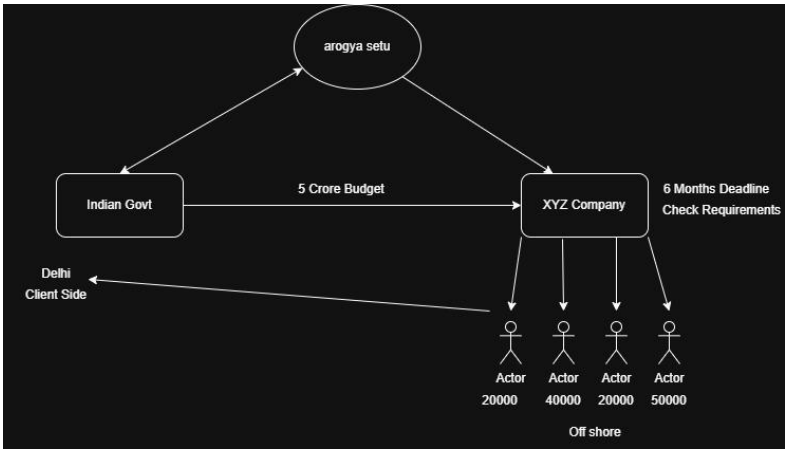
1. HR Department
2. sells Department
3. IT Department
4. marketing Department

IT COMPANY

1. Development team (java/python/c/react – Frontend/backend)
2. Tester Team (automation-selenium / manual)
3. DevOps Team (infra / db / app deploy to production env / exposed public)

TYPES OF COMPANY

1. service-based company
2. product-based company



WATER FALL METHOD

it is a method of following the step-by-step integration and deployment by manually.

1. The waterfall model is a sequential software development methodology.
2. It is a linear approach that follows a specific order of phases, starting from requirements gathering to maintenance.
3. Each phase must be completed before the next phase can begin.
4. The requirements and design phases are typically very detailed and time-consuming.
5. The testing phase comes after implementation, and any errors found during testing may require going back to previous phases to fix them.
6. Once a phase is completed, it is difficult and costly to make changes.
7. The final product is delivered after all phases are completed.
8. The waterfall model is less flexible compared to Agile or DevOps methodologies.
9. It is typically used for projects with well-defined requirements and a clear understanding of what the final product should look like.
10. The waterfall model is still used in some industries, such as construction or manufacturing.

Drawbacks

1. it’s a time consuming
2. difficult to change the code after it has been created.
3. without completing 1st step, we cannot go for 2nd step.
4. less feedback from the client.
5. high risk.
6. long development cycle.
7. difficult to manage complex projects.
8. lack of flexibility.

AGILE METHOD (it is successor of water fall method)

it is a build short and build often process.

1. Agile projects are completed in small increments or iterations, typically lasting a few weeks to a few months.
2. Teams work in close collaboration with each other and with stakeholders, such as customers and product owners.
3. Agile emphasizes frequent communication and feedback, with a focus on adapting to change rather than following a strict plan.
4. Agile relies on self-organizing teams that are empowered to make decisions and take ownership of their work.
5. Agile values working software over comprehensive documentation, although documentation is still important.
6. Agile methodologies include Scrum, Kanban, and Extreme Programming (XP), among others.
7. Agile can be used in a wide variety of industries and projects, from software development to marketing to construction.
8. Overall, Agile is an interative and flexible approach to project management that prioritizes collaboration, communication, and customer satisfaction.
9. Example: Jira

Drawbacks

1. limited documentation.
2. lack of predictability.
3. time consuming.
4. difficult for scaling, automation.
5. Dependency on Client Involvement.
6. Need for Skilled Team Members.

WHAT IS API (Application Programming Interface.)

it's a way for different software programs to talk to each other.

Think of an API like a waiter in a restaurant:

- You (the user/app) place an order
- The waiter (API) takes your request to the kitchen (server/system)
- The kitchen prepares the food

- The waiter brings the response back to you

You don’t need to know how the kitchen works—you just use the menu (API).

What APIs do

APIs let applications:

- Share data
- Request services
- Connect with other systems

Why APIs matter

- Save development time (no need to build everything from scratch)
- Allow integration between services
- Power most modern apps and websites

API METHODS

1. **GET (Read data) -** Used to fetch data from a server
2. **POST (Create data) -** Used to send data to server and create something new
3. **PUT (Update data) -** Used to update existing data (full update)
4. **DELETE (Remove data) -** Used to delete data

Real-life Example (Student example)

Imagine a student database API:

- GET → See student list
- POST → Add new student
- PUT → Update student info
- DELETE → Remove student

Real-life Example on Instagram

1. POST ----------> add a image on the Instagram
2. GET ----------> Read data of a post (like / comments) also scrolling reels
3. PUT/PATCH -----> update the post (add a song on post)
4. DELETE ----------> delete a post
